

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	23 JUNE 2026
Project Name	Metro Ticket Booking System in ServiceNow
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Metro Ticket Booking	Passenger enters booking details such as name, source station, destination station, journey date, passenger type, and number of tickets. Validate mandatory booking information before submission.
FR-2	Fare Calculation	Automatically calculate fare based on source station, destination station, passenger type, and number of tickets. Update total fare dynamically whenever ticket count or passenger type changes.
FR-3	QR Code Generation	Generate a dynamic QR code containing ticket information after the booking details are entered. Display the QR code on the booking page before confirming the booking.
FR-4	Ticket Confirmation	Save booking details into the u_metro_ticket_booking table using GlideAjax and Script Include. Generate a unique ticket record and display a booking confirmation message after successful submission.

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	The system should provide a simple, user-friendly interface for metro ticket booking with minimal user interaction.
NFR-2	Security	Only authenticated users can access the booking system, and booking data is securely stored in ServiceNow.
NFR-3	Reliability	The system should consistently calculate fares, generate QR codes, and save booking records without errors.
NFR-4	Performance	Fare calculation, QR code generation, and ticket booking should complete within a few seconds.
NFR-5	Availability	The application should be available whenever the ServiceNow instance is online to support continuous ticket booking.

NFR-6	Scalability	The system should support an increasing number of passengers, routes, and booking transactions without significant performance degradation.
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